KNOWLEDGE FACTOR—SOMETHING YOU KNOW, LIKE PASSWORDS

Knowledge factors are the most common—and weakest—authentication factors. These include passwords, PINs, and security questions. Knowledge factors are vulnerable to phishing, credential stuffing, and brute-force attacks. They can be lost, shared, or forgotten. Yet, despite these weaknesses, they are widely used because they are well understood and supported.

POSSESSION FACTOR—SOMETHING YOU HAVE, LIKE A SMARTPHONE

Possession factors nominally require that the requester physically possess an object. These objects include smartphones, hardware tokens, smart cards, and security keys. While possession factors are stronger than knowledge factors, they can still be lost, stolen, or cloned. In some cases, such as proving you have your phone or email account by receiving a text or email, they can be phished.

INHERENCE FACTOR—SOMETHING YOU ARE, LIKE BIOMETRICS

Inherence factors use unique biological traits to verify the requester's claim. These include things like fingerprints, facial recognition, retina scans, and voice recognition. Most people have probably used their face or fingerprint to unlock their phone, a prime example of using an inherence factor. However, storing biometric data remotely, such as in a central database, introduces significant security, privacy, and trust risks compared to storing it locally on a device. Unlike a password, if biometric data is stolen or fabricated, you can't change your face or fingerprint. Consequently, inherence factors are best used on local devices in combination with other factors. In addition, the rise of AI-generated *deepfakes* has provided the means to attack some biometric authentication systems.